

BrightPath Academy

Project Report

Submitted in partial fulfilment of the requirement for the award of

Degree of

BACHELOR OF COMPUTER APPLICATION

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CAREER POINT UNIVERSITY, KOTA

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Finally, I am grateful to everyone who directly or indirectly contributed to the successful completion of this project.

Bhrantik Nagar

Place: Career Point University,Kota

Date: 20/4/2026

DECLARATION

We hereby declare that this Project Report titled **BrightPath Academy** submitted by us and approved by our project guide, the School of Computer Application and Technology (SOCA), Career Point University, Kota, is a bonafide work undertaken by us and it is not submitted to any other University or Institution for the award of any degree diploma / certificate or published any time before.

Project Name : **BrightPath Academy**

Student Name: Bhrantik Nagar

Project Guide : Tanvi Thakur

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1 Project Title

BrightPath Academy – A Web-Based Learning Management System with Role-Based Access Control

2 Problem Statement

In the current digital learning environment, many beginners face difficulty in selecting the right course and following a structured learning path. Most platforms either lack proper guidance, organization, or user-friendly interfaces.

This project aims to solve these problems by providing a structured learning system with guided practice and role-based access for students and administrators. It allows users to easily register, explore courses, enroll, and access study materials while enabling administrators to manage users, enrollments, and courses efficiently.

3 Project Description

The BrightPath Academy system is a web-based learning management system designed to help users build digital skills through structured courses and guided practice. The system provides course listings, enrollment functionality, study materials, and an admin panel for managing the system efficiently.

3.1 Scope of the Work

In-Scope:

- User registration and login system
- Role-based access (Admin and Student)
- Viewing and selecting programs
- Course enrollment system
- Student dashboard with course details
- Study materials
- Admin panel for managing users, enrollments, and programs

Out of Scope:

- Online payment integration
- Live video streaming
- Third-party integrations

3.2 Project Modules

1. Home Module

Displays platform introduction and navigation options.

2. Programs Module

Shows available courses and learning tracks.

4. Student Dashboard Module

Displays enrolled course, tools, and study materials.

5. Study Material Module

Provides course content and practice tasks.

6. Admin Module

Allows management of users, enrollments, and programs including edit and delete operations.

3.3 Context Diagram (High Level)

Input:

- User registration details
- Login credentials
- Course selection
- Admin updates

Process:

- Authentication
- Enrollment processing
- Course and user management

Output:

- Dashboard information
- Course details
- Enrollment status

4 Implementation Methodology

The system is developed using a modular approach separating frontend and backend:

- Frontend handles user interface and interaction
- Backend manages authentication and database operations
- Role-based access ensures secure admin functionality
- Database stores user, course, and enrollment data
- CRUD operations are implemented for managing records

Testing includes:

- Form validation
- Login authentication
- Navigation testing
- Admin operations testing

5 Technologies to be used

5.1 Software Platform

- a) Front-end - HTML, CSS, JavaScript
- b) Back-end - Node.js, Express.js, Mysql

5.2 Hardware Platform

RAM: 4GB or above

OS: Windows 10/11

Browser: Chrome / Brave

Editor: VS Code

5.3 Tools, if any

- Visual Studio Code
- MySQL Workbench
- Github

6 Advantages of this Project

- Simple and user-friendly interface
- Role-based access control
- Efficient course and user management
- Structured learning system
- Scalable design

7 Assumptions, if any

None

8 Future Scope and further enhancement of the Project

- Addition of video lectures
- Progress tracking with analytics
- AI-based course recommendations

9 Definitions, Acronyms, and Abbreviations

Abbreviation	Description
CRUD	Create, Read, Update, Delete
JWT	JSON Web Token used for authentication

SQL	Structured Query Language
API	Application Programming Interface

10 Conclusion

Conclusion

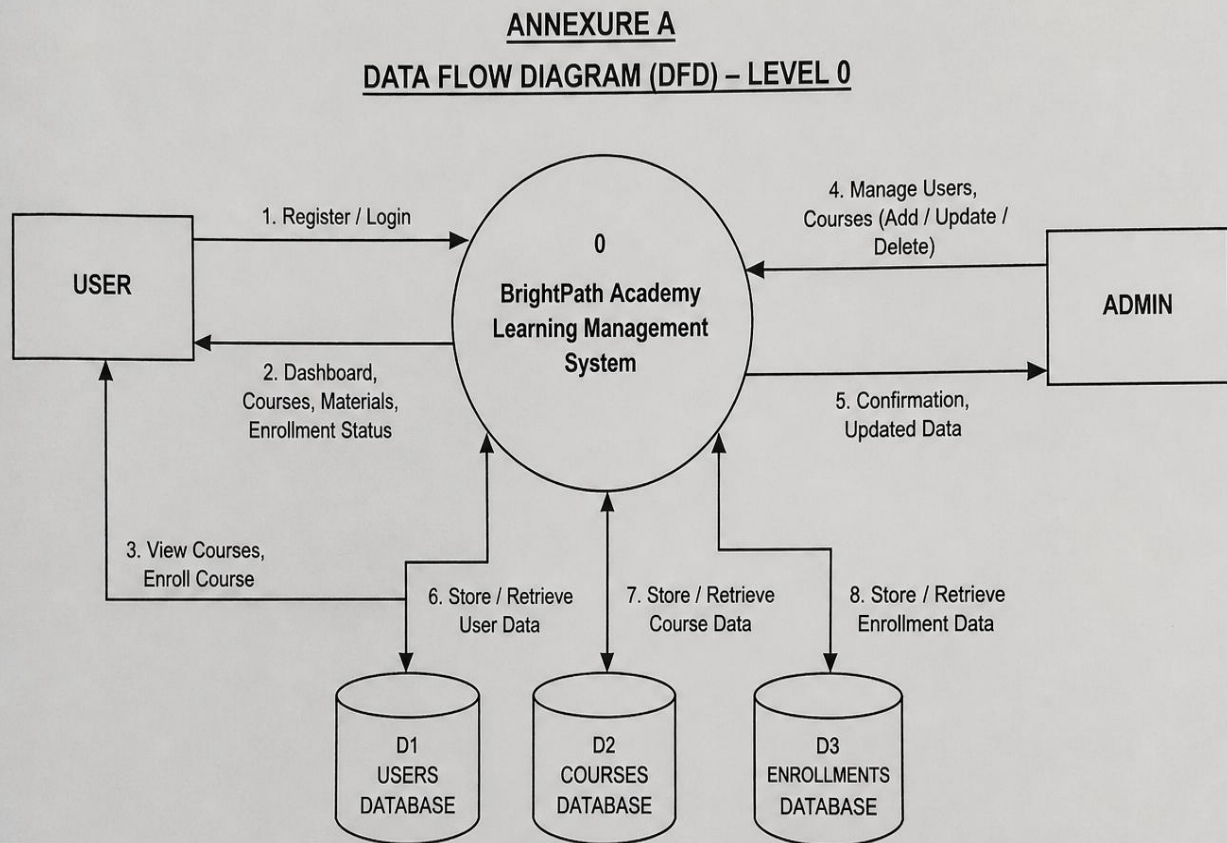
The BrightPath Academy system is a web-based learning platform designed to provide a simple and structured way for users to learn digital skills. It allows users to register, explore courses, enroll, and access study materials, while enabling administrators to manage users and courses efficiently.

The system reduces manual effort, improves data organization, and provides a user-friendly experience. It is cost-effective as it uses open-source technologies and can be easily maintained.

In future, the system can be enhanced by adding features such as online payments, video lectures, and progress tracking. Overall, the project demonstrates practical implementation of web development and database concepts in a real-world application.

Annexure A

Data Flow Diagram (DFD)



EXPLANATION:

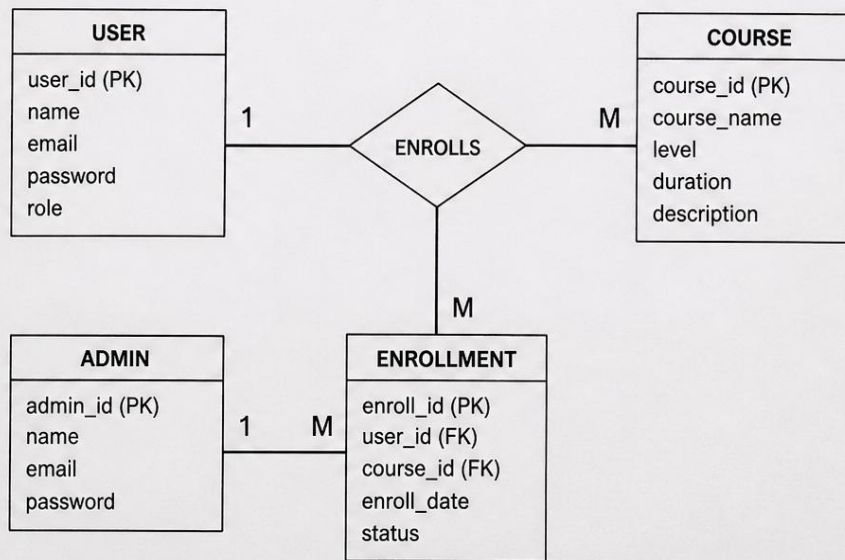
- User registers/logs in and views courses, enrolls in courses.
- Admin manages users and courses (add, update, delete).
- The system stores and retrieves data from Users, Courses and Enrollments databases.
- The system sends appropriate responses to User and Admin.

Annexure B

Entity-Relationship Diagram (ERD)

ANNEXURE B

ENTITY RELATIONSHIP DIAGRAM (ERD)



PK = Primary Key

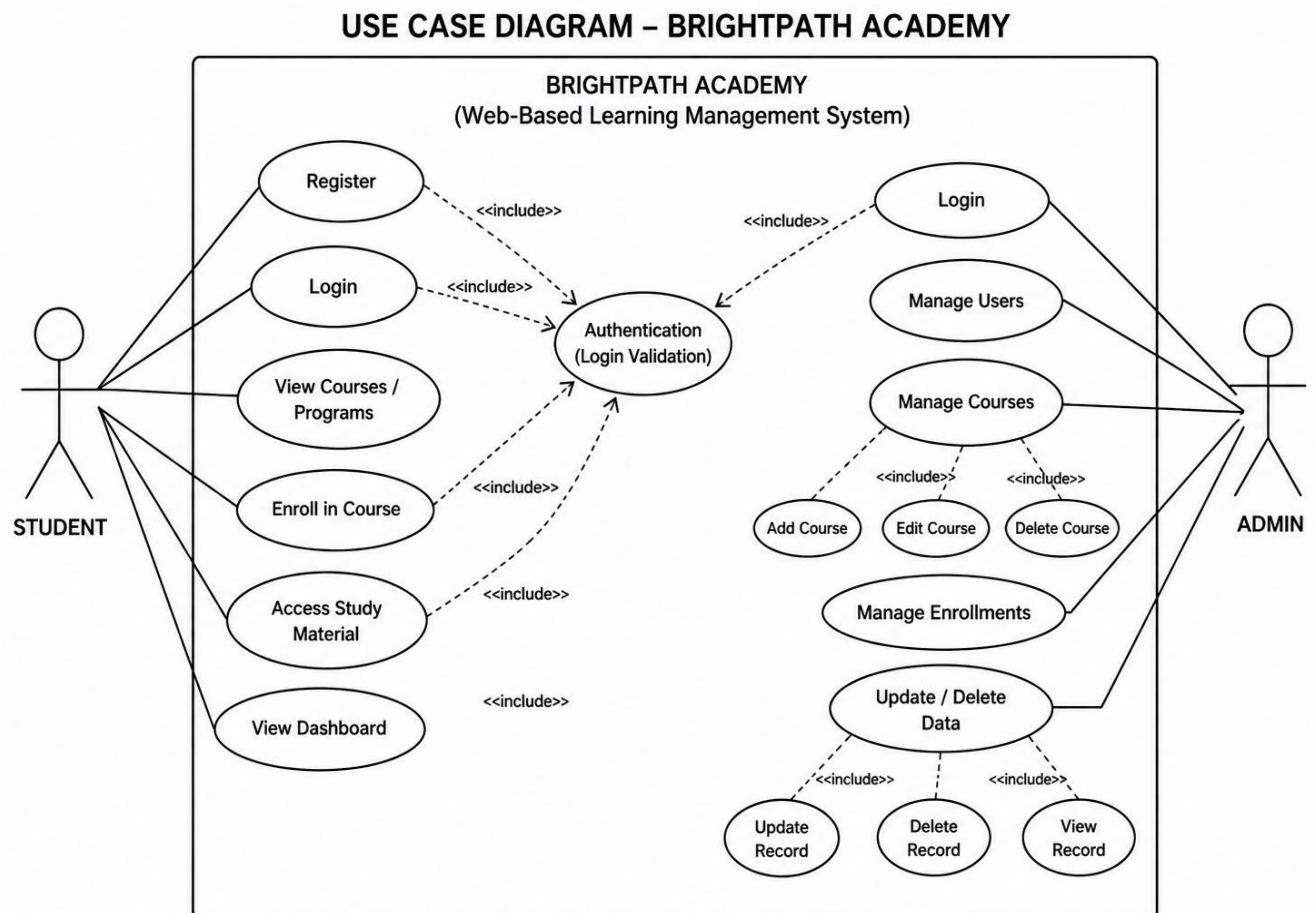
FK = Foreign Key

Explanation:

- A User can enroll in multiple Courses.
- A Course can have multiple Users.
- The Enrollment table stores the relationship between User and Course.
- Admin manages users and courses in the system.

Annexure C

Use-Case Diagram (UCD)



Annexure D

Data Dictionary (DD)

USER TABLE (USR):

Field Name	Data Type	Description
user_id	Number	Unique ID of the user
name	Text	Name of the user
email	Text	Email address
password	Text	Encrypted password
role	Text	User role (Admin/User)

COURSE TABLE (CRS):

Field Name	Data Type	Description
course_id	Number	Unique course ID
course_name	Text	Name of the course
level	Text	Course level
duration	Text	Duration of course
description	Text	Course details

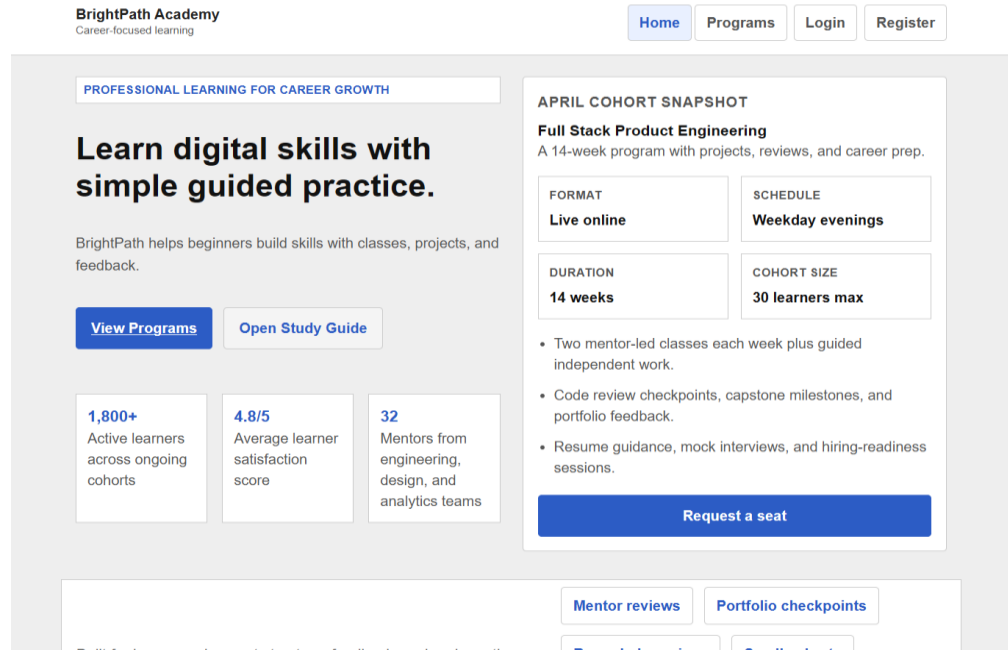
ENROLLMENT TABLE (ENR):

Field Name	Data Type	Description
enroll_id	Number	Unique enrollment ID
user_id	Number	Reference to user
course_id	Number	Reference to course
status	Text	Enrollment status (Pending/Active)

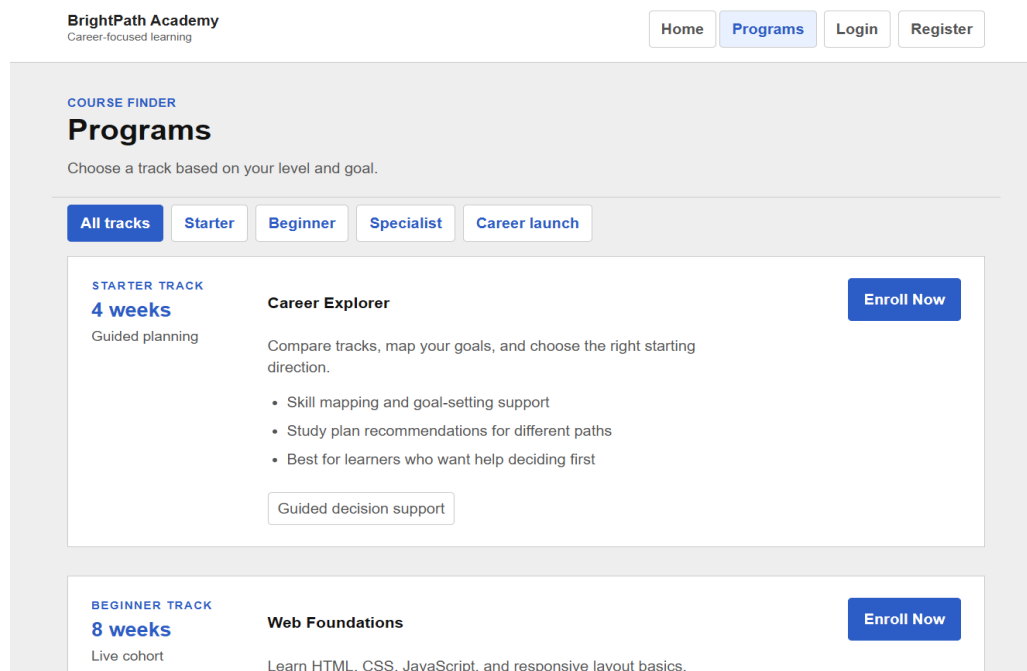
Annexure E

Screen Shots

Home Page:



Programes Page:



Login Page:

BrightPath Academy
Career-focused learning

HomeProgramsLoginRegister

STUDENT ACCESS
Sign in to your BrightPath account
Sign in to open your course and account.
Central dashboard
See your saved account profile, track focus area, and student role details.
SQL-backed auth
Your account data is stored in the backend.

LOGIN
Welcome back
Email Address
Password
Sign In
Need an account? [Create one here.](#)

Register Page:

BrightPath Academy
Career-focused learning

HomeProgramsLoginRegister

NEW ACCOUNT
Create your BrightPath profile
Create an account and save your learning track.
Organized structure
This area has its own backend and frontend structure.
Future-ready
It can be expanded later with more account features.

REGISTER
Start your student account
Full Name
Email Address
Preferred Learning Track
Password
Confirm Password
Create Account
Already registered? [Sign in here.](#)

User Dashboard Page:

BrightPath Academy
Career-focused learning

DashboardMy CourseLogout

STUDENT WORKSPACE
ansh, welcome back.
Your dashboard shows your course and account details.

student
Current role

Web Foundations
Selected track

2 May 2026
Member since

Open My Course

Open Web Foundations GuideLogout

Account details loaded successfully.

PROFILE SNAPSHOT
Your account details

Name
ansh

Email
ansh@gmail.com

Current course selection: Web Foundations

MY COURSE
Web Foundations
Build the web from first principles with guided practice

COURSE DETAILS
What you will work on
• Learn HTML, CSS, and JavaScript from scratch

Course Page:

Showing only the Web Foundations study material for this learner.

BEGINNER TRACK
Web Foundations
8 weeks - Beginner - Outcome: Junior Web Developer
Start with structure, styling, and interaction.
Modules

- HTML document structure, semantic tags, links, lists, tables, and forms.
- CSS selectors, box model, spacing, typography, color, flexbox, and grid.
- JavaScript variables, conditionals, loops, functions, and DOM selection.
- Mini projects, browser tools, file organization, and beginner Git practice.

Practice Tasks

- Build a semantic profile page.
- Create a responsive card layout.
- Make a button click counter with DOM events.

JAVASCRIPT PRACTICE
Button click counter
This shows how a button click can update the page.

STUDY CHECKLIST
Track your Web Foundations progress
Tick the boxes to update the progress

Admin Page:

ADMIN WORKSPACE

Admin access for lucky

Review activity, manage enrollments, and keep course records organized from one workspace.

12 USERS

4 ENROLLMENTS

3 INQUIRIES

3 NEWSLETTER

Admin data loaded.

LATEST ACTIVITY

Activity board

A cleaner pulse check on the newest learners, enrollments, questions, and signups.

U

ACCOUNTS

Recent Users

5

Newest registrations across learners and admins.

WEB FOUNDATIONS

ansh

ansh@gmail.com

Student

E

PIPELINE

Recent Enrollments

4

The latest course interest moving through the funnel.

23 APR 2026

test

Full Stack Product Engineering

Pending

Q

MESSAGES

Recent Inquiries

3

Questions that may need a follow-up from the team.

DATA ANALYTICS CAREER PATH

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Replied

Project Title: <BrightPath Academy>

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